

ADVAITH R PAI

Ernakulam, Kerala

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TECHNICAL SKILLS

Programming: Python, SQL, JavaScript
Tools & Platforms: Git, GitHub, Vercel
Coursework: Machine Learning, DBMS, Data Structures and Algorithms

EXPERIENCE

DRDO NPOL **September 2025 - October 2025**
Research Intern *Thrikkakara, Kochi*

- Developed hybrid **DWSTr (Depthwise Separable CNN + Transformer)** model for passive sonar classification, achieving **94.89% accuracy** across **12 vessel categories**
- Built end-to-end audio pipeline processing **10,000+ sonar samples** into Mel-spectrograms, reducing preprocessing latency by **35%**
- Improved model robustness using **SpecAugment**, **class-weighted loss**, and **6-layer Transformer encoder**, boosting minority recall by **18%** and F1-score by **12%**
- Optimized training with **early stopping and adaptive LR scheduling**, reducing training time by **25%** and implementing comprehensive performance evaluation dashboards
- **Tech Stack:** Python, TensorFlow/Keras, Librosa, NumPy, Scikit-learn, Matplotlib, Seaborn

PROJECTS

SERVD - AI-Powered Kitchen Assistant **February 2026**

- Developed full-stack AI recipe platform integrating **4+ external services/APIs (Gemini Vision AI, TheMealDB, Unsplash, Razorpay)** for automated ingredient recognition and recipe generation
- Implemented **2 pantry ingestion workflows** (image-based ingredient scanning + manual entry) with expiry tracking and real-time database synchronization
- Built secure authentication and subscription system with **tier-based premium access controls** and payment integration using Razorpay
- **Tech Stack:** Next.js, Tailwind CSS, Strapi, PostgreSQL/SQLite, Google Gemini AI, Clerk Auth, Razorpay, Arcjet

SELECTRA - AI Resume Analyzer **September 2025**

- Built AI-powered resume screening platform using **Claude Opus 4.1** and **PDF.js**, processing resumes to generate **ATS compatibility scores** and actionable improvement recommendations across 5 evaluation criteria
- Integrated **Puter.js cloud infrastructure** (file storage, KV database, AI API) with **Zustand** state management, enabling persistent resume tracking and **60% improved user retention**
- **Tech Stack:** React Router v7, TypeScript, Tailwind CSS, Puter.js, PDF.js, Zustand

EDUCATION & ACHIEVEMENTS

Muthoot Institute of Technology & Science
Bachelor of Technology in Computer Science - AI Engineering CGPA: 7.56/10

GitHub Leader: Gained **1000+ followers** through consistent contributions & high-quality projects

Open Source Consistency: Maintained **continuous GitHub activity** across **10+ months**, regularly shipping production-ready AI applications

Consistent Programming: Solved **300+ problems** on LeetCode, demonstrating strong Data Structures and Algorithms proficiency